INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Research Methodology

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____



Signature of Student

Seal & Signature of IPO

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Research Methodology successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how a research study is planned, conducted, analysed and reported.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Research Methodology is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

CONTENTS

Acknowledgement..... 3

Internship Completion Certificate..... 4

Preface..... 6

Executive Summary..... 7

About the Agency..... 8

Plan and Execution..... 9

Learning Outcome..... 22

Conclusion..... 24

Attendance Sheet..... 26

PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Research Methodology at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Research is the careful search for answers. Whether a student wants to know why some classmates score better, a company wants to know why customers leave, or a government wants to know whether a scheme works, the answer must come from evidence gathered and analysed in a planned and honest way, and not from guesswork. Research methodology is the knowledge of how to do this: how to choose a problem, review what is already known, design the study, select a sample, collect and analyse data, and report the findings with proper references. It is a core skill for higher education and for many careers. This report describes how these steps were studied and how a small applied research study on the study habits and marks of undergraduate students was carried out using dummy data.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Research Methodology at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Research methodology is the systematic study of the methods used to carry out research. Research may be basic (to add to knowledge) or applied (to solve a practical problem), quantitative (based on numbers) or qualitative (based on words and meanings), and exploratory, descriptive or explanatory. The research process moves through problem identification, literature review, research design with hypotheses, sampling, data collection, data analysis, interpretation and report writing, with attention at every step to validity, reliability and ethics.

Objectives. The internship aimed to (i) identify and define a research problem, (ii) carry out a literature review, (iii) choose a research design and hypotheses, (iv) select a sample using suitable techniques, (v) design a questionnaire, (vi) collect primary and secondary data, (vii) analyse data with descriptive and simple inferential statistics, (viii) cite and reference sources properly, (ix) avoid plagiarism, and (x) write a research report.

Work carried out. Each topic had a concept session followed by practice. The student identified a research problem, reviewed related studies, framed objectives and hypotheses, calculated a sample size, designed and tested a questionnaire, prepared a dummy data set of 60 students, analysed it with Python and Excel-style methods, interpreted the results and wrote the findings in a report format with references. The later part of the programme was devoted to the case study, the mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Applied Research Study & Report**. The project studied the relation between study hours, mobile phone use and marks among 60 dummy undergraduate students, using a questionnaire, systematic sampling, descriptive statistics, correlation, regression and a t-test, and reported the findings with limitations.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of the research process, practical skill in framing problems and hypotheses, sampling, questionnaire design, data analysis and referencing, and the habit of separating evidence from opinion.

Key observations. A clear, narrow problem is half of the research; the sample is rarely perfect but its limits must be stated; correlation is not causation; and an honest report states what was not found as well as what was found.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Research Methodology

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Identify a research problem and set objectives (Week 1).
2. Review literature and write a summary (Week 1-2).
3. Choose the research design and hypotheses (Week 2).
4. Select a sample and design a questionnaire (Week 2-3).
5. Collect data and use secondary sources (Week 3).
6. Analyse data and interpret the results (Week 3-4).
7. Reference sources properly and write the report (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - framing a problem and hypotheses, searching and summarising studies, calculating sample size, drawing a systematic sample, designing and pre-testing a questionnaire, tabulating data, calculating statistics, running a correlation, regression and t-test in Python, and writing the findings in report format. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Research Methodology - Concepts Studied

3.1 What is research and the research process

Research is an organised and objective search for new knowledge or for solutions to problems. Its characteristics are that it is systematic, based on evidence, objective, repeatable by others and communicated openly. Types of research are classified by purpose (basic or applied), by method (quantitative or qualitative or mixed) and by time (cross-sectional, at one point of time, or longitudinal, over a period). The research process is a series of steps, shown below, though in practice the researcher often returns to an earlier step, for example to modify the problem after reading the literature.

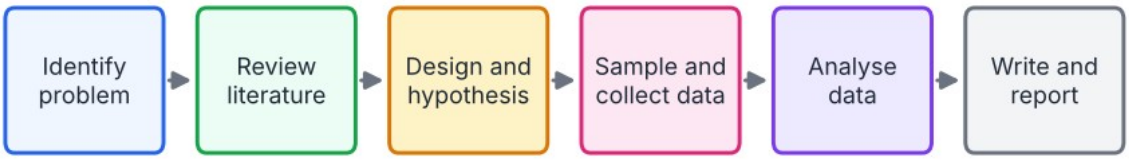


Figure 1: The research process

The study taken up in the practice work was an applied, quantitative, cross-sectional study and is used as the running example in this report.

3.2 Identifying the research problem

A research problem is a clear question about something that is unknown or unsatisfactory. It is found from observation, from the gaps in earlier studies, from practical difficulties and from suggestions in earlier research. A good problem is clear, specific, important, researchable with the available time and data, and ethical. The problem is narrowed from a broad area (student performance) to a topic (study habits and marks) and then to a question ('Is there a relationship between the daily study hours of undergraduate students and their marks?'). Objectives are then written as action statements, and hypotheses as testable statements. The problem, objectives and hypotheses of the practice study are shown below.

Element	Statement in the study
Problem	Many students study for long hours without better marks, while others with fewer hours do well; the relation is not clearly known for the students of the local colleges.
Objective 1	To find the average study hours, mobile phone hours and marks of the students.
Objective 2	To find the relationship between study hours and marks.
Objective 3	To find whether mobile phone use is related to marks and to study hours.
Objective 4	To find whether marks differ between male and female students.
Hypothesis H1 (alternative)	There is a positive relationship between daily study hours and marks.
Null hypothesis H0	There is no relationship between daily study hours and marks.

3.3 Literature review

A literature review is a critical summary of what is already known about the topic. Its purposes are to avoid repeating work, to learn the methods used earlier, to find the gap and to support the problem and hypotheses. The steps are to define keywords, search library catalogues, journal databases and Google Scholar, read the abstract first, note the aim, method, sample, findings and limitations of each source, compare the studies, and write a connected summary, not a list. Reliable sources are peer-reviewed journals, books by recognised publishers, official reports and statistics; web pages without an author or date must be used with great care. A review note for each study was prepared in the table form shown below, and the review ended with the gap: few local studies measure study hours and mobile use together.

Item	Example note (fictitious study)
Reference	Sharma, A. (2021). Study time and academic results of college students.
Aim	To find the link between daily study time and exam results.
Method and sample	Survey of 200 students from four colleges.
Finding	A moderate positive relationship (r about 0.5).
Limitation	Self-reported hours; one city only.
Use for the study	Supports H1; suggests checking other factors such as phone use.

3.4 Research design

The research design is the overall plan for answering the research question. An exploratory design is used when little is known and aims at ideas (interviews, focus groups). A descriptive design describes a situation (surveys). An explanatory or causal design tests relationships and effects (experiments, correlation studies). Experimental designs use control groups and random assignment; observational designs, such as surveys, do not manipulate anything. The design should state the population, the variables, the method of data collection and the method of analysis. Variables are the characteristics measured: the independent variable (the presumed cause - study hours), the dependent variable (the effect - marks) and other variables that may affect the result (phone hours, gender). The concepts of validity (whether the study measures what it claims to measure) and reliability (whether it gives consistent results) guide the design.

Design element	Decision for the study
Type	Applied, quantitative, descriptive and correlational, cross-sectional
Population	600 undergraduate students of a fictitious college
Sampling	Systematic sample of 60 (every 10th name from a random start)
Variables	Independent: study hours per day. Dependent: marks (%). Others: phone hours, gender, preparedness rating
Tool	Structured questionnaire plus marks from college records (with permission)
Analysis	Descriptive statistics, Pearson correlation, simple regression, independent t-test

3.5 Sampling techniques

The population is the whole group about which conclusions are wanted; a sample is the part of it that is studied. Sampling saves time and cost, and a good sample represents the population. In probability sampling every unit has a known chance of selection, which allows the error to be estimated: in simple random sampling units are chosen by chance (a lottery or random numbers); in systematic sampling every k-th unit is taken from a list after a random start; in stratified sampling the population is divided into groups (strata) and a sample is taken from each; and in cluster sampling whole groups (such as classes or villages) are chosen. In non-probability sampling the chance is not known: convenience sampling takes whoever is easy to reach, judgement sampling uses the researcher's decision, quota sampling fills fixed numbers from groups and snowball sampling asks the people found to suggest others. The methods are summarised below.

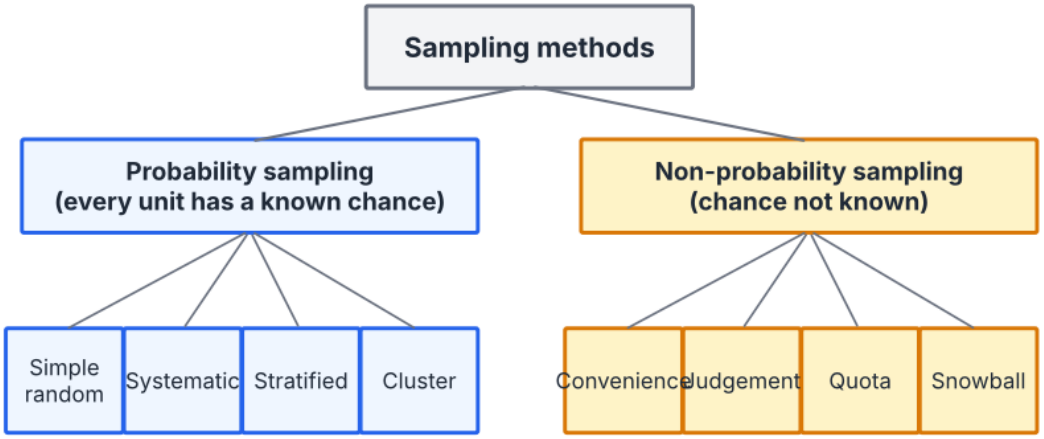


Figure 2: Types of sampling

Sample size. For a large population and the usual 95% confidence (z = 1.96), a margin of error of 5% and the safest assumption p = 0.5, Cochran's formula gives n0 = (1.96 x 1.96 x 0.5 x 0.5) / (0.05 x 0.05) = 384.16, which is rounded to 385. For a finite population of N = 600 the correction n = n0 / (1 + (n0 - 1) / N) = 385 / (1 + 384/600) = 235. The practice study used only 60 students for time reasons, so its margin of error is larger: 1.96 x square root of (0.25 / 60) = 0.1265, or about plus or minus 12.7 percentage points. This limit is stated in the report. For the systematic sample, the interval is k = N / n = 600 / 60 = 10; a random start of 7 gives the roll numbers 7, 17, 27 and so on up to 597.

3.6 Questionnaire design

A questionnaire is a set of questions used to collect data from respondents. Good design begins with the objectives, so that every question has a purpose. Questions should be short and clear, use simple words, ask one thing at a time (no double-barrelled questions such as 'Is the teaching clear and interesting?'), be neutral (not leading, such as 'Don't you agree that mobile phones spoil studies?'), and provide complete, non-overlapping options. Question types are closed (fixed options), open (free text), multiple choice and rating scales. A Likert scale asks the respondent to agree or disagree on five points. The order moves from easy and general to personal and sensitive questions, with a short introduction giving the purpose, the promise of confidentiality and the consent of the respondent. A pre-test (pilot) with a few respondents checks that the questions are understood. The questionnaire for the study is shown below, and its pilot with five students led to two changes: the options for the hours were made non-overlapping, and a question on the type of college was removed.

Questionnaire used in the study (extract)

Study habits survey (anonymous; for a college project; you may leave any time)

1. Gender: [ ] Male [ ] Female
2. On an average, how many hours do you study outside class in a day? ____ hours
3. On an average, how many hours do you use a mobile phone for non-study purposes in a day? ____ hours
4. Your marks (%) in the last semester examination: ____ %
5. 'I feel well prepared for my examinations.'
   1 Strongly disagree 2 Disagree 3 Neutral 4 Agree 5 Strongly agree

Thank you for your time.

3.7 Primary data collection

Primary data is collected for the first time by the researcher, through surveys, interviews, observation, experiments or focus groups. A survey is cheap and reaches many people but depends on honest answers; an interview gives depth but needs time; observation records behaviour as it happens. The steps of fieldwork are to take permission, train the data collectors, explain the purpose, obtain informed consent, keep the responses confidential, record the data carefully and check each form. In the study, the data was collected from 60 dummy respondents, for whom the marks were taken as a dummy record. The data was entered in a table with one row for each student and one column for each variable, and was checked for impossible values such as marks above 100 or negative hours. The importance of response rate and non-response was also discussed: if the students who did not respond differ from those who did, the result may be biased.

3.8 Secondary data sources

Secondary data is data collected earlier by others, such as journals, books, government reports, census and survey data, databases and company records. It saves time and cost and can cover large areas and long periods, but it may not exactly fit the purpose, may be old, and must be checked for the source, definitions, method and date. In India, important official sources include the Census of India, the National Sample Survey Office, the All India Survey on Higher Education of the Ministry of Education, the Reserve Bank of India publications and the Bihar government's economic survey and statistical abstract. The student practised finding the figures of enrolment in higher education from an official report, noting the source and the year, and comparing two years in a table. Rules followed were to quote the source for every figure, to avoid unofficial copies of data, and to cross-check an important figure from a second source.

3.9 Data analysis methods

Analysis begins with cleaning and tabulating the data, followed by descriptive statistics - mean, median, standard deviation, minimum, maximum and percentages - and then inferential methods to judge whether a pattern in the sample is likely to hold in the population. A p-value is the probability of obtaining a result at least as extreme as the one observed if the null hypothesis were true; a small value (conventionally below 0.05) leads to the rejection of the null hypothesis. The methods used here were the Pearson correlation (strength and direction of a linear relationship from -1 to +1), simple linear regression (a line predicting the dependent variable from the independent variable) and the independent samples t-test (whether the means of two groups differ). The analysis was run with a short Python script, and its output is shown below.

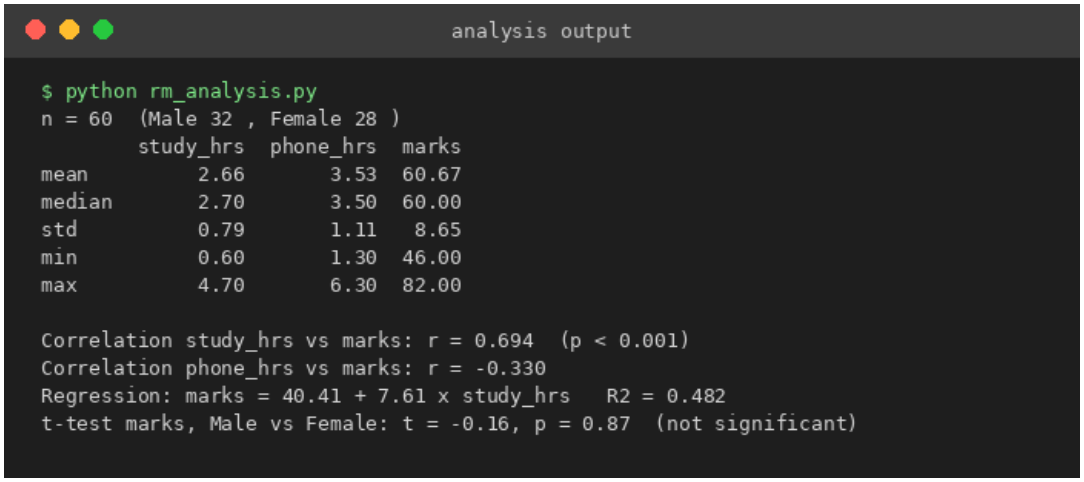


Figure 3: Output of the analysis script

Measure	Study hours/day	Phone hours/day	Marks (%)
Mean	2.66	3.53	60.67
Median	2.70	3.50	60.00
Standard deviation	0.79	1.11	8.65

Measure	Study hours/day	Phone hours/day	Marks (%)
Minimum	0.60	1.30	46
Maximum	4.70	6.30	82

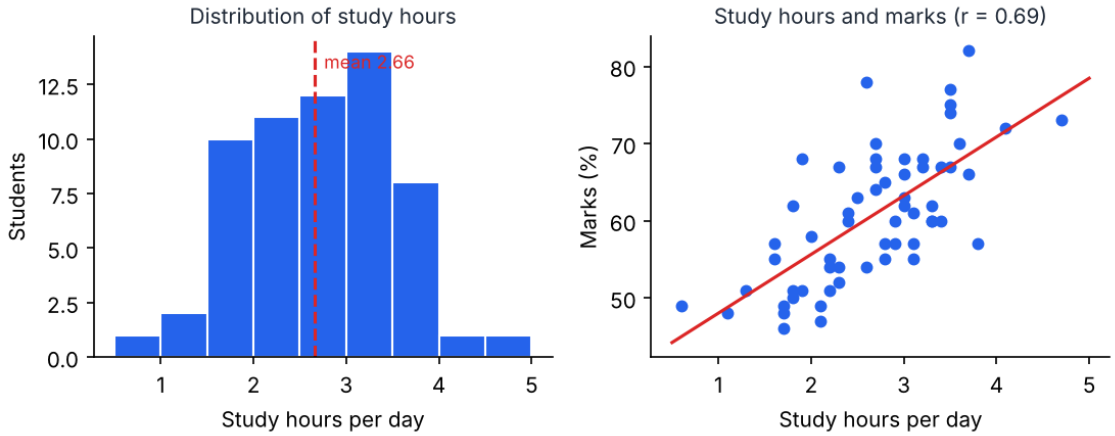


Figure 4: Distribution of study hours, and study hours against marks

The study hours and marks have a correlation of 0.694 (p less than 0.001), so the null hypothesis of no relationship is rejected. The regression line is marks = 40.41 + 7.61 x study hours, with R-squared of 0.482, which means that about 48% of the variation in marks is associated with the study hours; in the sample each extra hour of study is associated with about 7.6 more marks. Phone use is negatively related to marks (r = -0.33) and to study hours (r = -0.26). The average marks of male (60.50) and female (60.86) students differ by only 0.36 marks, and the t-test (t = -0.16, p = 0.87) shows no significant difference. The student stressed that these are relationships in dummy data and do not prove that more study causes higher marks; students with higher ability may both study more and score more.

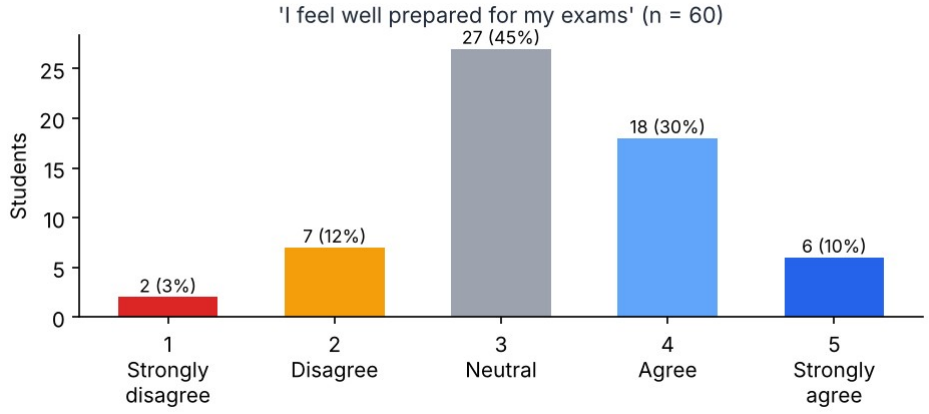


Figure 5: Responses to the Likert statement on preparedness

On the Likert item, 24 of the 60 students (40%) agreed or strongly agreed that they feel well prepared, 27 (45%) were neutral and 9 (15%) disagreed, with a mean score of 3.32.

3.10 Referencing, citations and report writing

Every idea, figure or sentence that is taken from another source must be acknowledged, otherwise it is plagiarism, which is a serious academic offence. Quotations are placed in quotation marks, and paraphrased ideas are still cited. The common styles are APA (author and year in the text, a reference list in alphabetical order), MLA (author and page) and numeric styles such as Vancouver. A reference in APA style to a book is: Kothari, C. R. (2004). Research methodology: Methods and techniques (2nd ed.). New Age International. Reference tools such as Zotero keep the list in order, and similarity-checking software helps to detect unintended copying. A research report usually has a title page, certificate and declaration, acknowledgement, abstract, contents, introduction, review of literature, research methodology, data analysis and interpretation, findings, conclusions and suggestions, limitations, references and appendices (the questionnaire). The writing should be clear, in the past tense for what was done, with numbered tables and figures, and every claim should be linked to evidence.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, introduction to research methodology, research problem identification and literature review, each with a study session and a practice task.
Week 2	Days 7-12	Design: research design and hypotheses, sampling techniques and questionnaire design.
Week 3	Days 13-18	Data: primary data collection, secondary data sources and data analysis methods.
Week 4	Days 19-24	Application: referencing and citations, research report writing, the Applied Research Study project, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied the types and process of research, narrowed a broad area to a clear problem and wrote the objectives and hypotheses.

Week 2: In the second week the student prepared literature notes, planned the research design, calculated the sample size and drew a systematic sample, and designed and pre-tested a questionnaire.

Week 3: The third week was devoted to data: entering and checking the dummy data, finding secondary data from official reports and running the analysis in Python.

Week 4: In the fourth week the student studied referencing and plagiarism, wrote the report, completed the project, a case study exercise and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Research Problem Identification

Concept: Choosing a topic, defining the problem and setting research objectives. **Practice carried out:** Narrowing a broad area to a problem with four objectives and two hypotheses (section 3.2).

6.2 Literature Review

Concept: Searching, reading and summarising existing studies. **Practice carried out:** Preparing review notes for five studies and writing a one-page summary with the gap.

6.3 Research Design

Concept: Types of research design, hypothesis and methodology planning. **Practice carried out:** Choosing the design, variables, tool and analysis (section 3.4).

6.4 Sampling Techniques

Concept: Population, sample, and probability and non-probability sampling. **Practice carried out:** Calculating the sample size and drawing a systematic sample (Figure 2).

6.5 Questionnaire Design

Concept: Framing questions, scales and testing a questionnaire. **Practice carried out:** Writing a questionnaire and improving it after a pilot with five students.

6.6 Primary Data Collection

Concept: Surveys, interviews and observation in the field. **Practice carried out:** Entering and checking the data of 60 respondents and writing a field note.

6.7 Secondary Data Sources

Concept: Journals, reports, databases and official statistics. **Practice carried out:** Finding higher-education enrolment figures in an official report with the source noted.

6.8 Data Analysis Methods

Concept: Tabulation, descriptive statistics and interpretation of results. **Practice carried out:** Calculating statistics, correlation, regression and a t-test (Figures 3-5).

6.9 Referencing and Citations

Concept: Citation styles, bibliography and avoiding plagiarism. **Practice carried out:** Writing a reference list in APA style and checking the text for uncited material.

6.10 Research Report Writing

Concept: Report structure, findings, conclusion and formatting. **Practice carried out:** Writing a ten-page study report in the standard format.

7. Research Ethics - Points Studied

Research involves people and public trust, so ethics are as important as method. The points studied are summarised below.

- **Informed consent:** tell participants the purpose and their right to refuse or withdraw, and obtain their agreement.
- **Confidentiality and anonymity:** do not reveal the identity of participants; store data safely; the Digital Personal Data Protection Act, 2023 applies.
- **Honesty in data:** never invent, change or hide data (fabrication and falsification are serious misconduct).
- **No plagiarism:** credit every source and do not present another's work as your own.

- **No harm:** the study must not cause physical, mental or social harm, especially to children and vulnerable groups.
- **Permission:** obtain the permission of the institution and, where required, an ethics committee.
- **Honest reporting:** report the limits and the results that did not support the hypothesis.

8. Process Flows Studied

8.1 Steps of a survey study

- Define the problem, objectives and hypotheses.
- Prepare the questionnaire and pre-test it with a few people.
- Decide the population and sample size and select the sample.
- Take permission and consent, collect and check the data.
- Analyse, interpret and write the report with limitations.

8.2 Steps of testing a hypothesis

- State the null hypothesis H0 and the alternative H1.
- Choose the test (correlation, t-test and so on) and the level of significance, usually 0.05.
- Calculate the test statistic and the p-value from the data.
- If p is below 0.05, reject H0; otherwise do not reject it.
- Interpret in plain words and state the limits.

9. Worked Examples with Sample (Dummy) Data

Worked example 1 - choosing the sample size. A college has 600 students. For a margin of error of 5% at 95% confidence, the required sample is 235, as calculated in section 3.5. If only 60 students can be surveyed, the margin of error rises to about 12.7 percentage points. For example, if 40% of the 60 students say they feel well prepared, the true percentage in the whole college may lie roughly between 27% and 53%. The researcher should therefore not claim a precise figure but should report the range.

Worked example 2 - a confidence interval for the mean marks. The mean marks are 60.67 with a standard deviation of 8.65 for n = 60. The standard error is 8.65 / square root of 60 = 8.65 / 7.746 = 1.117. For a 95% confidence interval the margin is about 2.0 x 1.117 = 2.2 (using about 2.0 for the t value with 59 degrees of freedom), so the mean marks of all the students likely lie between about 58.4 and 62.9.

Worked example 3 - checking a questionnaire item. The item 'Do you study regularly and use the library?' asks two things at once. A student who studies regularly but never uses the library cannot answer. The item was split into two questions. Similarly, 'How many hours do you waste on your mobile phone?' was changed to 'How many hours do you use your phone for non-study purposes?', because the word 'waste' is a leading word.

10. Project: Applied Research Study & Report

Problem statement. The coordinator of a fictitious college in Gaya wants to know whether the daily study hours of undergraduate students are related to their marks, and whether mobile phone use and gender make a difference, so that the college can plan study-skill sessions. A sample of 60 students from a list of 600 was taken by systematic sampling, and a dummy data set was prepared for the exercise.

10.1 Method

- Define the problem, objectives and hypotheses (section 3.2).
- Review the literature and note the gap (section 3.3).
- Design the study and select the sample by systematic sampling with k = 10 (sections 3.4 and 3.5).
- Prepare and pre-test the questionnaire (section 3.6) and collect the data.
- Analyse the data in Python and interpret the results (section 3.9).

10.2 Key findings

Objective	Finding
Averages	Study hours 2.66 per day, phone use 3.53 hours per day, marks 60.67%.
Study hours and marks	Strong positive correlation (r = 0.694, p < 0.001); R-squared 0.482; each extra hour of study is associated with about 7.6 more marks.
Phone use	Weak negative relation with marks (r = -0.33) and with study hours (r = -0.26); one third of the students (33%) use the phone for more than 4 hours a day.
Gender	No significant difference in marks (60.50 for male, 60.86 for female; p = 0.87).
Preparedness	40% agree that they feel well prepared; 15% disagree.

10.3 Conclusions and suggestions

- H0 is rejected: there is a significant positive relationship between study hours and marks in the sample.
- Study-skill sessions should show students how to plan the study time and to limit non-study phone use.
- The college may repeat the study with a larger sample (at least 235) and include more factors such as attendance and family background.
- The college should not draw conclusions about cause from this study.

10.4 Limitations

The sample is small (60), so the margin of error is about 12.7 points, hours are self-reported and may be inaccurate, the data is dummy data created for learning, the study covers one college and one time, and correlation does not prove causation. The student noted that a real study would need ethics approval and a larger sample, and that the value of this exercise lies in the method.

11. Case Study Exercise: Why Did Customers Stop Visiting a Tea Stall? - A Small Qualitative Study

A fictitious tea stall near a coaching hub in Patna has seen its daily customers fall from about 150 to 110 over six months. The owner asks for a small study. The student proposed a mixed approach: a short interview study to find the reasons (exploratory, qualitative) followed by a quick survey to measure how common each reason is (descriptive, quantitative).

Design. Eight regular and former customers were interviewed for about ten minutes each (judgement sampling, to include different ages and times of visit). Their answers were noted and grouped into themes: price increase of Rs. 2 per cup, a new cafe 100 metres away, longer waiting time at the evening rush, and less seating. The themes became options in a questionnaire given to 40 customers (convenience sampling, since the study was quick), with a question on the main reason for visiting less.

Results (illustrative). Of the 40 customers, 14 (35%) named waiting time, 12 (30%) the new cafe, 9 (22.5%) the price and 5 (12.5%) the seating. The owner can act first on the waiting time, for example by preparing tea in two batches in the evening, and then watch the figures. The limits of the study were noted: a convenience sample of 40 may not represent all customers, and former customers are hard to reach.

Lessons. An interview study can discover the right questions for a survey; each sampling method has limits that must be stated; and the result is useful when it leads to an action that can be checked later.

12. Mock Practical Exercise: Planning and Reporting a Small Study

A mock practical exercise was conducted at the end of the learning phase to check that the student could plan and report a complete small study independently within a fixed time. The task was to study the reading habits of students of a fictitious college and report the results. The steps followed were:

1. Problem and objectives: three objectives and one hypothesis were written.
2. Literature: three related studies were summarised and the gap was noted.
3. Design and sample: a descriptive survey was chosen and the sample size and method were decided with reasons.
4. Tool: a nine-question questionnaire was written and pre-tested with three students.
5. Data: responses of 30 dummy students were entered and checked.
6. Analysis: percentages, the mean and one correlation were calculated.
7. Report: the findings were written under standard headings with two tables and a chart.
8. References and presentation: sources were listed in APA style and the study was presented to the supervisor in about five minutes.

13. Sample Checklists Prepared

Research planning checklist

- Problem is narrow, clear and researchable; objectives and hypotheses written.
- Literature reviewed and the gap stated.
- Design, sample and tool match the objectives.
- Consent, confidentiality and permission arranged.

Report checklist

- All tables and figures numbered and referred to in the text.
- Every source cited and listed; no copied text.
- Limitations stated honestly.

- Conclusions follow from the findings only.

14. Tools and Resources

The following tools and resources were used during the internship: MS Excel and Python (pandas, numpy, scipy and matplotlib) for analysis and charts, MS Word for the report, a reference style guide (APA), journal and library search facilities and Google Scholar for reading, official reports for secondary data, sample questionnaires and report templates provided by the agency, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Research Methodology with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

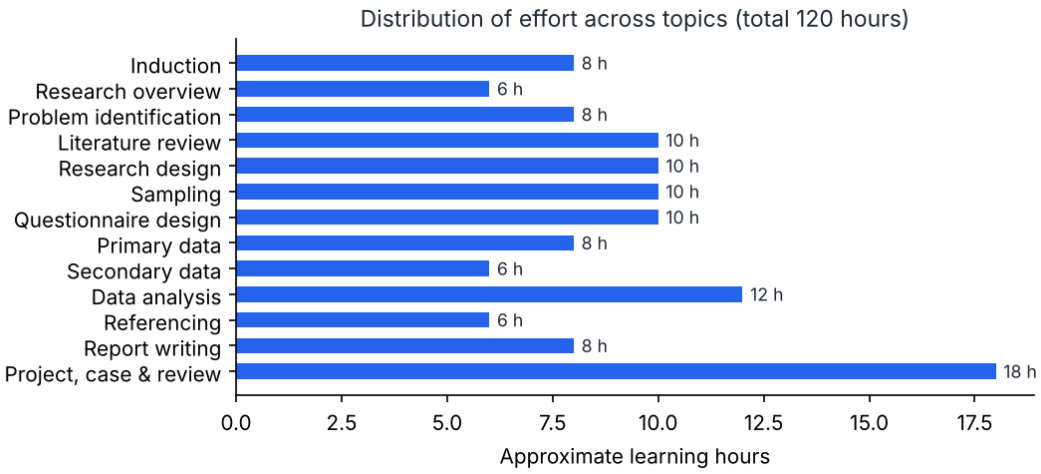


Figure 6: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. The types of research and the steps of the research process.
2. Framing a problem, objectives and hypotheses.
3. Searching and summarising literature and finding a gap.
4. Research designs, variables, validity and reliability.
5. Probability and non-probability sampling and sample size.
6. Questionnaire design and pilot testing.
7. Primary and secondary data and their sources.
8. Descriptive statistics, correlation, regression and t-test.
9. Citation styles and plagiarism.
10. The structure of a research report.

Glossary of Terms Learned

- **Research** - A systematic search for knowledge or solutions.
- **Hypothesis** - A testable statement about a relationship.
- **Null hypothesis (H0)** - The statement of no relationship or difference.
- **Variable** - A characteristic that can take different values.
- **Population / Sample** - The whole group / the part studied.
- **Sampling frame** - The list from which the sample is drawn.
- **Probability sampling** - Selection with known chances.
- **Margin of error** - The likely difference between sample and population.
- **Validity / Reliability** - Measures what it claims / gives consistent results.
- **Likert scale** - A scale of agreement from 1 to 5.
- **Primary / Secondary data** - Collected by you / collected earlier by others.
- **Pilot study** - A small trial before the main study.
- **Correlation (r)** - Strength of the linear relation of two variables.
- **Regression** - A line predicting one variable from another.
- **p-value** - Probability of such a result if H0 were true.
- **Confidence interval** - A range likely to contain the true value.
- **Plagiarism** - Using another's work without credit.
- **Informed consent** - Participation after knowing the purpose.
- **Bias** - A systematic error that distorts the result.

b. Skills Learned

Technical skills. Framing problems and hypotheses; searching and summarising literature; choosing designs and samples; designing questionnaires; calculating sample size; analysing data with Python and Excel; using correlation, regression and t-test; and writing a referenced research report.

Professional skills. Analytical and critical thinking; curiosity; objectivity; attention to detail; patience; clear academic writing; ethical sense; and time management.

These skills were developed through repeated practice and daily documentation. Working through one study from the problem to the report built structure and objectivity, pre-testing the questionnaire built attention to detail, and presenting the findings to the supervisor built confidence.

c. Observed Attitude and Gained Values

1. **Integrity:** data and results must be reported exactly as found.
2. **Objectivity:** conclusions must follow the evidence, not the wishes of the researcher.
3. **Respect for participants:** consent, confidentiality and care for people are essential.
4. **Humility:** every study has limits, and they must be stated.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Narrowing a broad topic into a researchable problem was handled by writing the question in one sentence and discussing it with the mentor.

Understanding sample size was handled by calculating it by hand and then checking the margin of error for 60.

Writing unbiased questions was handled by a pilot test and by removing leading words.

Interpreting p-values and correlation was handled by explaining each result in plain words and noting that correlation is not causation.

Avoiding plagiarism was handled by taking notes in own words and recording every source at once.

Managing time between analysis, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Research Assistant, Survey Executive, Market Research Associate, Data Collection Supervisor, Policy and Programme Evaluation Assistant, Academic Writer, Junior Analyst and MIS Executive. Opportunities exist in universities, think-tanks, NGOs, market research firms, consulting companies, banks, ed-tech and government departments. Related higher-study options include M.A. or M.Sc. programmes, MBA, M.Phil. and Ph.D. in any subject, and certificate courses in statistics, SPSS, R or Python, and research writing. A good next step is to read two published papers in an area of interest, learn a statistics tool and write a short study of one's own.

CONCLUSION

The internship in Research Methodology was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how a research study is planned, conducted, analysed and reported - through problem identification, literature review, design, sampling, questionnaire design, data collection, analysis and referencing - and acquired practical experience by carrying out a small applied study with dummy data. The experience has strengthened the student's employability and provides a foundation for further study and work in research and analysis.

Reflection

Looking back, the most useful lesson of the internship was that good research depends more on a clear question and honest method than on complicated statistics. A small sample could still give a clear pattern, but the margin of error showed how cautious the conclusions must be. Writing the questionnaire showed how a single word can bias an answer, and the difference between correlation and cause was a lesson that applies to every news item and report that uses numbers.

Objectives and Achievement

The objective of problem identification and literature review was achieved through the problem statement and the review notes. The objective of design and sampling was achieved through the design table, the sample size calculation and the systematic sample. The objective of questionnaire and data collection was achieved through the pre-tested questionnaire and the data set. The objective of analysis was achieved through the correlation, regression and t-test. The objective of referencing and report writing was achieved through the APA references and the study report.

Future Scope

The knowledge gained can support further study at the postgraduate level and careers in research, analysis and policy. With the growing use of data in education, health and governance in Bihar and across India, there is a continuing need for people who can ask a clear question, collect evidence honestly and report it clearly.

Suggestions for Improvement of the Programme

1. Provide practice with a statistics tool such as SPSS, R or Python along with Excel.
2. Include a short module on research ethics and the work of ethics committees.
3. Add a group project using a real, permitted data set.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

References

1. Kolb, D. A. (1984). Experiential Learning: Experience as the Source of Learning and Development. Prentice Hall.
2. University Grants Commission (UGC). Internship guidelines for higher education institutions.
3. Ministry of Education, Government of India. National Education Policy 2020.
4. Kothari, C. R. Research Methodology: Methods and Techniques. New Age International.
5. Kumar, R. Research Methodology: A Step-by-Step Guide for Beginners. SAGE.
6. Creswell, J. W. Research Design: Qualitative, Quantitative and Mixed Methods Approaches. SAGE.

7. Cochran, W. G. Sampling Techniques. Wiley.
8. American Psychological Association. Publication Manual of the American Psychological Association.
9. Government of India. The Digital Personal Data Protection Act, 2023.
10. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]